1 Write a Python program that demonstrates variables, data types, conditional statements, and loops using a real-life example

AIM

To write a Python program that demonstrates:

- Variables (store student details and marks)
- Data Types (string, integer, float, list)
- Conditional Statements (decide grade based on percentage)
- Loops (iterate through subjects to calculate total marks)

ALGORITHM

1. Start the program.
2. Create variables to store student name, subjects, and marks.
3. Use a loop to calculate the total marks.
4. Compute percentage = (total marks / maximum marks) × 100.
5. Use conditional statements to assign grades:
   - ≥ 90 → Grade A
   - ≥ 75 → Grade B
   - ≥ 50 → Grade C
   - Else → Grade D
6. Display student details, marks, percentage, and grade.
7. End program.

CODE

```
# Student Grade Calculator


# Variables and Data Types
student_name = "Yogin"                    # string
subjects = ["Math", "Science", "English"]  # list of strings
marks = [85, 92, 78]                      # list of integers
```

```python
max_marks = 100                    # integer (per subject)

# Display student info
print("Student Name:", student_name)
print("Subjects and Marks:")

# Loop to display marks
total = 0
for i in range(len(subjects)):
    print(f"{subjects[i]}: {marks[i]}/{max_marks}")
    total += marks[i]

# Calculate percentage
percentage = (total / (max_marks * len(subjects))) * 100
print("\nTotal Marks:", total)
print("Percentage:", percentage)

# Conditional Statements for Grade
if percentage >= 90:
    grade = "A"
elif percentage >= 75:
    grade = "B"
elif percentage >= 50:
    grade = "C"
else:
    grade = "D"

print("Grade:", grade)
```

OUTPUT

Student Name: Yogin

Subjects and Marks:

Math: 85/100

Science: 92/100

English: 78/100

Total Marks: 255

Percentage: 85.0

Grade: B

CONCLUSION

This program successfully demonstrates:

- Variables (, , , , , )
- Data Types (string, integer, float, list)
- Conditional Statements (grading logic)
- Loops (iterating through subjects and marks)

2 Write a program to check whether a given number is prime or not using functions

AIM

To write a Python program that uses functions to check whether a given number is prime or not.

ALGORITHM

1. Start the program.
2. Define a function  that:

- Returns  if  (since prime numbers are greater than 1).
- Loops from  to  to check divisibility.

- If divisible, return .
- Otherwise, return .

3. In the main program, take a number as input.
4. Call the function  and display whether the number is prime or not.
5. End program.

CODE

```
# Function to check prime number
def is_prime(n):
    if n <= 1:
        return False
    for i in range(2, int(n**0.5) + 1):  # loop till sqrt(n)
        if n % i == 0:
            return False
    return True

# Main Program
num = int(input("Enter a number: "))

if is_prime(num):
    print(num, "is a Prime Number")
else:
    print(num, "is NOT a Prime Number")
```

OUTPUT

```
Enter a number: 7
7 is a Prime Number
```

CONCLUSION

This program demonstrates:

- Functions ( encapsulates logic).
- Conditional Statements (checking divisibility and prime conditions).
- Loops (iterating through possible divisors).
- A real-life application of prime numbers in cryptography, coding, and mathematics.